Dynamic RL-Based LLM Execution Graph Partitioning

Final report: dataset pipeline, v5r statistical reanalysis, the v6 policy framework, and the pre-registered v7 main run

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Repository	github.com/MahaprabhuK/Dynamic-RL-LLM-Execution-Graph-Partitioning
Working branch	`improve/meaningful-partitioning` (main has the original v1 pipeline)
Model	TinyLlama/TinyLlama-1.1B-Chat-v1.0, float32, CPU only
Hardware	Laptop, 10 physical / 16 logical CPU cores, 16 GB RAM (RTX 3050 present but unused; torch is a CPU build)

Abstract

On a single CPU machine, the fastest number of threads for LLM inference depends on how busy the machine is, and that can change while a response is being generated. We test whether re-deciding the thread budget every 10 tokens helps, using TinyLlama-1.1B (fp32) on a 10-core / 16-thread laptop with synthetic background load. In a pre-registered run of 450 requests of 200 tokens (10 prompts × 5 load scenarios × 9 conditions), a lookup table built from a load sweep on the same machine (`rule_lookup`) was 0.1% to 4.8% faster than the best fixed thread budget in each scenario, chosen in hindsight from the same data, and faster in all 10 contexts when load arrived mid-request. Against the fixed `balanced` and `threads_2` arms it was up to 13% and 20% faster; 15 of 20 pre-registered paired t-tests were significant after Holm correction. An online bandit warm-started from the same sweep nearly tied it. A DQN trained offline on the sweep was 10% to 60% slower than `rule_lookup` in four of five scenarios. 106 of 450 runs had paging events; removing them keeps every effect direction. Policies built from direct measurements of the machine were the most reliable.

> **Statistical correction (12 October 2026).** The original v5 prose used latency-only outlier exclusions and unadjusted interval-based significance labels. The source CSV has no independent paging telemetry. The v5r reanalysis in `data/processed/adaptive_report_v5r.txt` retains all 40 contexts, reports paired t and Wilcoxon tests with Holm-adjusted p-values, and supersedes the original v5 inferential claims below. Earlier v3/v4 significance labels are historical and have not been recomputed here.

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1. Summary

The project builds a scheduler that decides, while an LLM response is being generated on a single CPU machine, how many threads the model should use. TinyLlama-1.1B (fp32) runs layer by layer through a hand-written decode loop with one shared KV cache, so the thread budget can change between tokens without changing the output: token IDs are identical across all arms. In the final setup (v7, section 16) the arms are single-stage budgets of 2, 3, 4, 6 and 10 threads (`threads_2` … `threads_10`) plus `balanced` (two sequential stages of 3 threads). Four fixed arms (`threads_2`, `balanced`, `latency_optimized` = 6 threads, `static` = 10 threads) are compared with five policies that re-decide every 10 tokens from measured external CPU load: `rule_lookup` (a lookup table from a load sweep on this machine), `cores_heuristic` (10 minus the busy logical CPUs, not fitted to data), `rl_adaptive` (a DQN trained offline on the same sweep), `bandit_adaptive` (an online bandit warm-started from the sweep) and `rule_v5` (the earlier 60% threshold rule). Versions v3-v5 used three arms, `static` (10 threads), `latency_optimized` (6) and `balanced` (3+3); they are described as history in sections 4-11.

The work went through eight phases. The early versions (v1/v2) produced results that turned out to be noise, because the strategies were computationally identical and were measured one after another. v3 fixed both problems: strategies differ in thread budget, and measurements are paired and randomized under controlled background CPU load. The project then moved to its main idea, **re-deciding the thread budget every 10 tokens during generation**, tested with a DQN and an online bandit in v3-v5, and finally in a pre-registered main run (v7) with a wider set of arms and policies.

v7 main run (pre-registered, 450 runs; section 16)

- **Against the best fixed arm in each scenario** (chosen from the same data; descriptive), the lookup rule built from the v7 load sweep (`rule_lookup`) was 0.1% to 4.8% faster. The largest gain was in spike, where it was faster in all 10 contexts; in steady_idle it tied (`experiments/e4_best_fixed.csv`).
- **Pre-registered tests** (paired t, Holm over 20): `rule_lookup` beat fixed `threads_2` in all five scenarios (-8% to -20% vs threads_2) and fixed `balanced` in steady_idle, spike and relief (-13%, -5%, -9% vs balanced). Against `balanced` in steady_heavy and ramp (-2%), the differences did not survive Holm correction. 15 of 20 tests were significant.
- **Policies built on direct measurements of this machine beat the offline DQN.** The online bandit, warm-started from the same sweep, nearly tied `rule_lookup` (descriptive). The DQN did not generalise across prompts: it picked the best arm in 35% of held-out contexts offline, and online it was 10% to 60% slower than `rule_lookup` in four of five scenarios. The core-count heuristic was significantly slower in three scenarios.
- 106 of 450 runs had a paging event. With those removed (n = 3-9 per test) the direction of every primary effect is unchanged, but fewer tests survive Holm correction.

Earlier results (v3-v5)

- Which strategy is fastest depends on load. Idle machine: 10 threads (`static`) wins. Around 8 busy workers: 6 threads. At 10-14 busy workers: 3+3 threads (`balanced`). At 12 workers `static` oversubscribes the CPU and is about 18x slower per token than `balanced` (2.14 s vs 0.116 s per token).
- In all five scenarios, the re-deciding DQN (`rl_adaptive`) and the 60% threshold rule (`rule_adaptive`) are statistically indistinguishable (Holm-adjusted p = 1.0, mean differences -0.6 to +12.5 s). The gain comes from intra-generation re-deciding, not from reinforcement learning.
- Compared to the best fixed baseline, `rl_adaptive` did not show a statistically significant advantage in any scenario after Holm adjustment across the 30 predefined comparisons. In ramp (all contexts, n=8), `rl_adaptive` was 16% slower than fixed `balanced` (68.9 s vs 59.4 s, mean diff +9.52 s, Holm p = 1.0). In spike, adaptive policies lost 21.8 s to fixed `balanced` (59.5 s vs 37.7 s, Holm p = 0.44) due to reaction delay.
- For `rl_adaptive` versus `rl_once` in ramp, the all-context mean difference was -238.40 s (95% paired bootstrap CI [-366.99, -101.04]; raw paired-t p = 0.0134). After Holm adjustment across the 30 comparisons, Holm p = 0.31, which does not survive the 0.05 family-wise significance threshold.
- For `rl_adaptive`, the only comparisons that survived Holm adjustment in the all-context primary analysis were against `static` in ramp (-353.7 s, Holm p = 0.0009) and spike (-73.7 s, Holm p = 0.0268), reflecting the severe oversubscription cliff of 10 threads under external load. Across all 30 tests in the family, the only other comparison that survived Holm adjustment was `rule_adaptive` versus `balanced` in relief (-3.66 s, Holm p = 0.0177).
- Token IDs remained identical across all seven conditions in all 40 v5 contexts. The source data has no per-segment counterfactual action outcomes, so an offline segment oracle cannot be computed on v5.

2. Problem and goal

LLM inference on a shared machine competes with other processes for CPU. A single fixed thread budget is only best for one load level. Too many threads under contention cause oversubscription and context switching. Too few threads on an idle machine waste cores. The goal is a scheduler that observes the request and the system state and picks the execution plan (thread budgets {3, 6, 10} applied via layer staging) that minimizes latency, without changing the model's output.

Most schedulers decide once per request, before the first token. A long response can take tens of seconds, and load can change during that time, so the per-request decision goes stale. The core method evaluated here is **intra-generation re-scheduling**. Because every strategy shares the same weights and the same KV cache, the engine can switch thread budgets between tokens without recomputation overhead and with identical output.

3. Project timeline

Date	Commit	Work
26 Jul 2026	60ef0b8	Initial project structure
1 Aug 2026	b08ba1d	Dataset pipeline (Alpaca download and preprocessing to `prompts.csv`), TinyLlama inference benchmark
2 Aug 2026	14f631e	LLM inference pipeline, benchmarking, runtime telemetry (CPU/RAM)
15 Sep 2026	34e2043	RL-based execution graph scheduling: DQN agent, environment, execution graph (networkx), static scheduler (current `main`)
10 Oct 2026	ef2f544	Snapshot of the v2 layer-sliced engine and its results, stop tracking `.pyc` files
10 Oct 2026	5291ab2	v3: strategies differ by thread budgets, paired randomized protocol, contextual-bandit framing, tests, config loader, README
10 Oct 2026	3f42bdb	Intra-generation re-scheduling, external-load monitor, dynamic load, v3 dataset, pilot, DQN v3, first adaptive run
11 Oct 2026	6d5a62b	v4: online contextual bandit, non-blocking telemetry, token-triggered load, resume and report-only modes
11 Oct 2026	3c2fc89	Robust bandit (stall rejection, bin hysteresis), load sweep, ramp scenario
11-12 Oct 2026	this commit	v5 run finished overnight (40 contexts x 7 conditions), `--keep-awake` flag, v5 results, this report
13 Oct 2026	e9a9fad - ecec569	v7 engine (arms 2-10 threads, segment and system-paging telemetry, RAM guard), v7 load sweep (288 runs), `rule_lookup`, reaction-delay pilot, DQN v7, bandit prior, pre-registered main-run plan; main run starts (67 runs)
17-18 Oct 2026	4d6896e - 0a8e3b2	Machine fingerprint, CPU capacity and sanity gates; power-plan fixes; plan amended to prompts 21-30 (450 runs)
18-19 Oct 2026	this commit	v7 main run finished (450/450), final checks, analysis (section 16)

4. System design

4.1 Layer-sliced inference engine

`src/llm/inference.py` runs TinyLlama without `model.generate()`. It embeds the prompt, then runs the decoder layers stage by stage with a shared KV cache, applies the final norm and LM head, and samples greedily. Layer staging is the execution mechanism: before each stage it sets `torch.set_num_threads()` to that stage's intra-op budget. Because all stages execute sequentially on the CPU, partitioning serves as an intra-request thread budgeting mechanism rather than parallel pipeline parallelism. Between stages it records an activation handoff (an in-process `clone()`, timed and sized). The model is loaded lazily and once.

`generate_partitioned_response(prompt, strategy, scheduler=..., decision_interval=N)` accepts a scheduler object. Every N tokens it calls `scheduler.decide()` and switches strategy when asked. All runs in v4/v5 generate exactly 200 tokens (`ignore_eos`), so latencies are comparable across conditions.

4.2 Execution strategies

The action space consists of three thread budgets: {3, 6, 10} threads. Layer staging applies these budgets across the 22 decoder layers:

ID	Strategy	Thread budget	Layer stages	Thread allocation per stage	Execution behavior
0	`static`	10	0-21 (single stage)	10 threads	Runs all 22 layers at 10 threads
1	`balanced`	3	0-10, 11-21 (two stages)	3 threads, 3 threads	Runs layers 0-10 at 3 threads, then layers 11-21 at 3 threads sequentially (3 threads continuous)
2	`latency_optimized`	6	0-21 (single stage)	6 threads	Runs all 22 layers at 6 threads

In `balanced`, stage 2 executes only after stage 1 completes. Because the two stages run sequentially one after the other at 3 threads each, in effect `balanced` runs at 3 threads throughout generation. Strategies are defined in `configs/config.yaml` and loaded by `src/config.py`. `src/llm/partition.py` turns them into stage definitions, and `src/graph/execution_graph.py` builds a networkx execution graph annotated with stage layer ranges and thread counts.

4.3 Telemetry and load generation

- `ExternalLoadMonitor` measures system CPU minus this process's own CPU, so the scheduler reacts to other processes and not to its own inference threads.
- `BackgroundLoadSampler` (v4) runs that measurement in a daemon thread with smoothing. Reading it never blocks generation. In v3, blocking telemetry cost 0.21 s per request.
- `BackgroundCPULoad` starts N busy-loop worker processes for a whole request. `DynamicCPULoad` changes N over time. `TokenScheduledLoad` (v4) changes N after a given number of generated tokens, so every policy meets the load change at exactly the same point of the response.

4.4 Schedulers and policies

Condition	What it does
`static`, `balanced`, `latency_optimized`	Fixed strategy for the whole request (baselines)
`rule_adaptive`	Threshold rule on external CPU (60% in v5): below it use `static`, above it `balanced`. Re-decided every 10 tokens with hysteresis
`rl_once`	Offline DQN, decides once before the first token
`rl_adaptive`	Same DQN, re-decides every 10 tokens with hysteresis
`bandit_adaptive`	Online contextual bandit (`OnlineBanditPolicy`) that learns from measured seconds per token while it serves

Hysteresis (`patience=2`) means a policy only switches after it proposes the same new strategy in two decisions in a row. This stops flapping on noisy readings, but it also means a switch takes at least 20 tokens after the load changes.

4.5 The online bandit

Each 10-token segment is one bandit round. The context is the external-CPU bin, the arm is the strategy, and the cost is the measured seconds per token of the segment just generated (the first segment, which contains prefill, is skipped). Features:

- Warm start from the offline DQN's Q-values with 2 pseudo-observations per arm.
- Discounted estimates (discount 0.98 per round in a bin) so stale data fades when the machine drifts.
- UCB-style exploration bonus (0.08) so arms not tried recently are re-checked, at the price of one segment.
- Stall rejection (v4.1): a segment more than 2x slower than the best estimate in its bin is ignored, unless 5 arrive in a row, in which case the bin is reset and relearned.
- Bin hysteresis: the bin only changes within a request when the reading moves more than `bin_margin` points past an edge (v5: edges 51% and 63%, margin 4).
- One instance keeps learning across all requests. Its state is saved to `adaptive_bandit_state_*.json` so interrupted runs can resume.

4.6 Offline RL component

The DQN (`src/rl/dqn_agent.py`) is an MLP with two hidden layers of 64 units, input `[prompt_length, input_tokens, cpu%, ram%]` and one Q-value per strategy. Since requests are independent, it is trained as a contextual bandit with `gamma=0`. The logged data contains the outcome of every strategy in every context, so all three rewards are learned per context (full feedback). The reward is negative seconds per generated token. CPU and RAM penalties exist in `src/rl/reward.py` but are off by default, because post-run snapshots were too noisy. Features are built by one shared function (`src/rl/features.py`) for training and serving.

5. Phase 1 (v1/v2): first pipeline and why its results were noise

The `main` branch has the first full pipeline: Alpaca prompts, TinyLlama inference, CPU/RAM telemetry, a DQN scheduler and evaluation scripts. The v2 work then added the layer-sliced engine. Results from that stage:

Run	static	balanced	latency_optimized	Note
v1 fixed strategies, mean latency (20 prompts)	14.83 s	25.14 s	22.60 s	Sequential runs
v2 fixed strategies, mean latency (20 prompts)	24.08 s	22.73 s	22.36 s	Sequential runs
v2 RL policy evaluation	25.12 s	12.45 s	19.74 s	RL picked `latency_optimized` 20/20 times, 19.37 s

The rankings flip between runs. Review of the code showed why: in v1/v2 the three strategies did the same computation with the same threads, only grouped differently. All runs of one strategy were also done before the next one started, so system drift (other apps, thermal state, memory pressure) was confounded with the strategy. The differences were noise. These files are kept as `*_v2.csv` for reference and are not used for conclusions.

6. Phase 2 (v3): real strategies and a corrected methodology

Commit 5291ab2 changed the strategies so they execute differently by enforcing thread budgets {3, 6, 10} across stages (with balanced running two sequential stages at 3 threads each, still fp32 and token-identical). It also changed the measurement protocol. Strategies are now run back to back in random order within the same context (prompt, repeat, load level), after warm-up, under a controlled number of background busy

workers. The state is measured before the decision, real generated token counts are used, and results come with bootstrap 95% CIs and paired differences.

6.1 Pilot (3 prompts, 0/4/8 workers)

Workers	static (10 thr)	balanced (3+3)	latency_opt (6)	Paired result vs static
0	6.70 s	8.26 s	8.17 s	legacy report marked balanced slower; not Holm-adjusted
4	20.98 s	18.19 s	17.27 s	legacy report marked latency_opt faster; not Holm-adjusted
8	72.63 s	41.95 s	28.13 s	legacy report marked both alternatives faster; not Holm-adjusted

This was the first evidence that the strategies have real, load-dependent winners.

6.2 v3 dataset (10 prompts x 5 load levels x 3 strategies = 150 runs, 50 tokens each)

Workers	static	balanced	latency_opt	balanced vs static (paired)	latency_opt vs static (paired)
0	8.70 s	11.27 s	9.35 s	+30.1% (legacy report label, unadjusted)	+11.8% (legacy report label, unadjusted)
2	8.42 s	8.40 s	8.20 s	+6.1% (legacy report label, unadjusted)	+0.6% (legacy report label, unadjusted)
4	19.42 s	12.81 s	16.62 s	-27.3% (legacy report label, unadjusted)	-16.2% (legacy report label, unadjusted)
6	24.71 s	15.94 s	22.40 s	-26.4% (legacy report label, unadjusted)	-17.7% (legacy report label, unadjusted)
8	44.27 s	22.87 s	45.32 s	-34.9% (legacy report label, unadjusted)	+65.8% (legacy report label, unadjusted)

Conclusion from v3: `static` wins on an idle machine, `balanced` wins from 4 busy workers upward. The threshold baseline was set to static/balanced from these numbers.

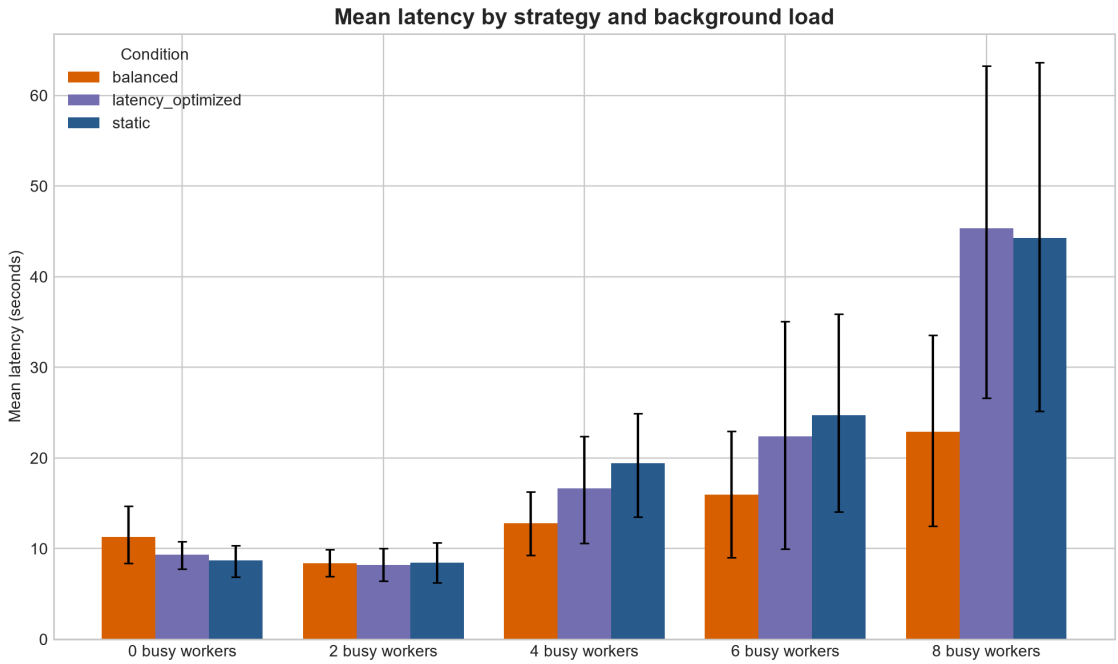


Figure 1. v3 mean latency per strategy and background load (bootstrap 95% CIs).

7. Phase 3: contextual-bandit DQN

The DQN was trained for 200 episodes on the v3 data with full feedback, with prompts split into train and held-out validation sets. On the 15 held-out contexts it had zero regret in 6 of them and a mean regret of 0.049 s per token against the oracle (best strategy in hindsight). It learned the main pattern in the data: mostly `static` at 0-2 workers and `balanced` at 4-8 workers. The model is saved as `data/processed/dqn_model_v3.pth` and is the offline policy used in all later runs.

8. Phase 4: intra-generation re-scheduling (v3 adaptive)

Commit 3f42bdb added re-scheduling during generation and `evaluate_adaptive.py`. Four scenarios: `steady_idle`, `steady_heavy`, `spike` (load arrives mid-request) and `relief` (load leaves mid-request). Six conditions were compared on 60-token requests with time-triggered load, 3 prompts per scenario.

The legacy v3 analysis reported no clear benefit for intra-generation scheduling and a 35% descriptive slowdown versus fixed `latency_optimized` in `spike`. Its small sample and earlier unadjusted statistical procedure do not support a current significance claim. Short requests, time-triggered rather than token-triggered load, blocking telemetry, and possible system drift were proposed as explanations, not isolated causal findings.

9. Phase 5 (v4): online bandit

v4 setup: 10 prompts x 4 scenarios = 40 contexts, 7 conditions, 200 tokens per request, decision every 10 tokens, load triggered by token count, non-blocking telemetry. The historical report excluded two latency-outlier candidates, but the v4 raw file contains no page-fault or swap counters; the paging label is unverified and all raw data is retained.

Scenario	balanced	bandit	latency_opt	rl_adaptive	rl_once	rule_adaptive	static
relief	20.98	19.09	17.96	19.68	20.61	19.62	20.16
spike	24.75	22.69	22.70	24.04	24.63	24.86	24.17
steady_heavy	25.21	24.20	23.27	25.13	25.16	25.25	27.33
steady_idle	19.60	17.29	17.18	17.52	17.77	17.46	17.27

Mean latency in seconds for a 200-token response. v4 heavy load was 8 workers.

- The historical report observed mean differences of -7.8% in `spike` and -7.0% in `relief` versus the per-request DQN. Its significance labels were not adjusted across comparisons and have not been recomputed; treat these as descriptive only.
- It corrected a stale offline model. The v3 data said `balanced` wins under load. At v4 time the 6-thread `latency_optimized` won everywhere. The DQN kept choosing `balanced`; the bandit moved to `latency_optimized`.
- Regret against the best fixed strategy in hindsight: 0.3-1.2 s per request for the bandit, 0.9-2.1 s for the DQN. Bandit regret fell from 0.94 s to 0.66 s between the first and second half of the run, so it was learning.
- Limitation: one fixed strategy happened to be near-optimal everywhere, so the bandit only tied it in `spike` and `steady_idle` and was 4-6% slower in `relief` and `steady_heavy`. That is the cost of exploration (about 3 switches per request).
- Decision overhead fell from 0.21 s (v3) to 0.002 s per request.

v4 diagnostics motivated changes to the bandit: unusually slow segments were rejected under a latency-based rule, and many switches happened at readings near a bin edge (the count quoted in earlier versions of this report cannot be reproduced from the committed CSVs, which record 121 bandit switches in v4). The available CSV does not prove those slow segments were paging stalls, nor that bin-edge noise caused every counted switch. The resulting implementation added outlier rejection, bin hysteresis, slower forgetting (discount 0.98), and a smaller exploration bonus (0.08).

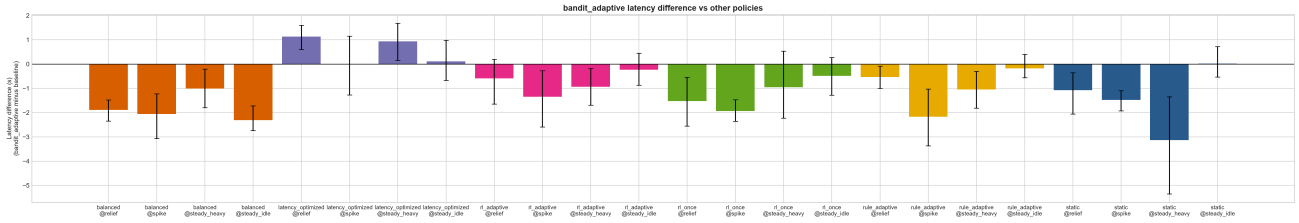


Figure 2. v4 paired differences of the online bandit against every other condition (negative = bandit faster).

10. Phase 6: load sweep

Because v4 had only one winning strategy, `pilot_load_sweep.py` measured decode seconds per token for every strategy at 0-14 busy workers (2 repeats) to find load levels where the winner changes.

Workers	static (10 thr)	latency_opt (6)	balanced (3+3)	Best
0	0.0790	0.0793	0.0879	static
2	0.0799	0.0892	0.0899	static
4	0.0827	0.0917	0.0998	static
6	0.0930	0.0938	0.1053	static
8	0.1099	0.0965	0.1072	latency_optimized
10	0.1223	0.1135	0.1087	balanced
12	2.1352	0.1510	0.1157	balanced (static 18x slower)
14	3.0548	0.8520	0.2154	balanced

Decode seconds per token, mean of 2 repeats (`data/processed/load_sweep_v4.csv`). External CPU readings: 7-49% at 0-6 workers, 54-60% at 8, 66-89% at 10-14.

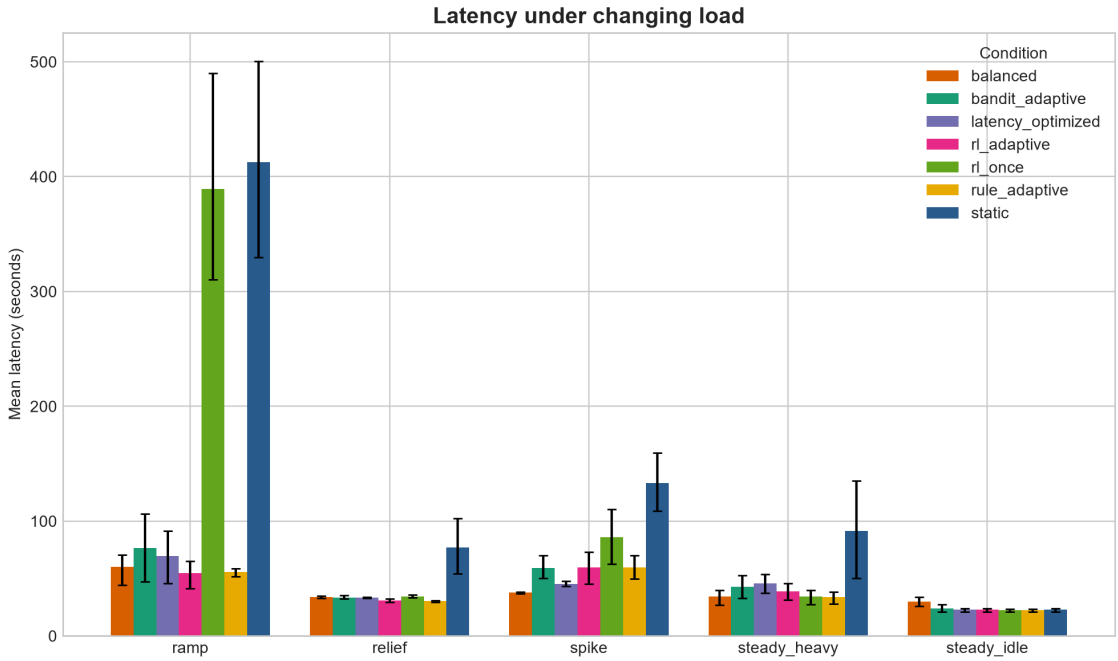
The sweep shows three distinct thread-budget regimes and an oversubscription cliff. Up to 6 workers, all thread budgets are within about 15% of each other. At 8 workers, 6 threads (`latency_optimized`) wins. At 10-14 workers, 3 threads (`balanced`, running 3+3 threads sequentially) wins. At 12 workers, 10 threads (`static`) collapses because 10 inference threads plus 12 busy processes oversubscribe 16 logical cores (~18x slower per token than 3 threads). These regimes set the v5 bandit bin edges (51%, 63%) and the rule threshold (60%).

11. Phase 7 (v5): multi-level load and the ramp scenario

v5 setup: 8 prompts (prompt IDs 21-28) x 5 scenarios = 40 contexts, 7 conditions, 280 runs of 200 tokens. v5r reanalyses all source rows and retains all 40 contexts. The original report's four latency-outlier flags (IDs 14, 19, 36, 39) are preserved only as sensitivity candidates: `adaptive_runs_v5.csv` records no page faults, paging, or swap counters, so paging cannot be independently confirmed. Every context has all seven conditions and identical token IDs. Mean decision overhead recorded by the original run was 0.005 s per request.

11.1 Mean latency (seconds, 200 tokens)

Scenario	balanced	bandit	latency_opt	rl_adaptive	rl_once	rule_adaptive	static
ramp (n=8)	59.4	84.9	71.7	68.9	307.3	**56.4**	422.6
relief (n=8)	33.8	33.9	33.4	32.2	34.3	**30.2**	91.7
spike (n=8)	**37.7**	59.2	45.3	59.5	86.0	59.4	133.2
steady_heavy (n=8)	34.8	43.5	48.0	44.4	35.3	**34.8**	93.5
steady_idle (n=8)	29.7	23.7	22.8	22.8	**22.4**	22.5	22.7



Legacy v5 plot, generated from the old latency-only exclusion analysis. The all-context v5r values are shown in the table and report file.

11.2 Comparisons against best fixed strategy and rule baseline

The primary evaluation leads with comparisons against the best fixed strategy in hindsight and against the 60% threshold rule (`rule_adaptive`). Deciding once on an idle machine is a weak baseline, because 10 threads under external load is a known oversubscription cliff.

Scenario	Comparison	Mean diff (s)	95% CI (s)	Raw p (t-test)	Holm p (30 tests)	Verdict
ramp	rl_adaptive vs balanced (best fixed)	+9.52 s	[-6.28, 39.02]	0.5328	1.0000	not significant (rl_adaptive 16% slower)
ramp	rl_adaptive vs rule_adaptive	+12.52 s	[-7.47, 42.35]	0.4151	1.0000	not significant
ramp	rule_adaptive vs balanced	-3.00 s	[-10.59, 6.26]	0.5400	1.0000	not significant
relief	rl_adaptive vs latency_opt (best fixed)	-1.15 s	[-3.87, 2.52]	0.5529	1.0000	not significant
relief	rl_adaptive vs rule_adaptive	+2.04 s	[-0.36, 5.51]	0.2672	1.0000	not significant
relief	rule_adaptive vs balanced	-3.66 s	[-4.88, -2.59]	0.0006	0.0177	significant (favours rule_adaptive)
spike	rl_adaptive vs balanced (best fixed)	+21.80 s	[7.58, 35.54]	0.0200	0.4405	not significant after Holm adjustment
spike	rl_adaptive vs rule_adaptive	+0.05 s	[-14.57, 15.63]	0.9953	1.0000	not significant
spike	rule_adaptive vs balanced	+21.75 s	[11.94, 32.14]	0.0057	0.1433	not significant after Holm adjustment
steady_heavy	rl_adaptive vs balanced (best fixed)	+9.60 s	[-0.05, 22.27]	0.1717	1.0000	not significant
steady_heavy	rl_adaptive vs rule_adaptive	+9.68 s	[0.18, 21.38]	0.1552	1.0000	not significant
steady_heavy	rule_adaptive vs balanced	-0.08 s	[-1.34, 0.94]	0.8972	1.0000	not significant
steady_idle	rl_adaptive vs static (best fixed)	+0.05 s	[-0.20, 0.32]	0.7164	1.0000	not significant
steady_idle	rl_adaptive vs rule_adaptive	+0.27 s	[-0.12, 0.69]	0.2773	1.0000	not significant
steady_idle	rule_adaptive vs balanced	-7.21 s	[-10.57, -4.83]	0.0030	0.0792	not significant after Holm adjustment

DQN vs threshold rule: Across all five scenarios, the re-deciding DQN (`rl_adaptive`) and the 60% threshold rule (`rule_adaptive`) are statistically indistinguishable (Holm p = 1.0, mean differences -0.6 to +12.5 s). The gain comes from intra-generation re-deciding, not from reinforcement learning.

11.3 Comparison against deciding once (rl_once)

Scenario	rl_adaptive vs rl_once	95% CI (s)	Raw p (t-test)	Holm p (30 tests)	Verdict
ramp	-238.40 s	[-366.99, -101.04]	0.0134	0.3073	not significant after Holm adjustment
relief	-2.10 s	[-4.76, 1.52]	0.2742	1.0000	not significant
spike	-26.46 s	[-55.34, 3.49]	0.1476	1.0000	not significant
steady_heavy	+9.14 s	[-0.03, 21.03]	0.1700	1.0000	not significant
steady_idle	+0.44 s	[0.10, 0.82]	0.0676	1.0000	not significant

Deciding once with `static` (10 threads) on an idle machine is a weak baseline because 10 inference threads plus 12 workers on 16 logical cores creates a severe oversubscription cliff when load arrives. In ramp, deciding once took 307.3 s on average (n=8). Although the raw paired t-test yields p = 0.0134, it does not survive Holm adjustment across the 30 comparisons (Holm p = 0.31). Across the entire primary analysis of 30 comparisons, the only tests that survived Holm adjustment were `rl_adaptive vs static` in ramp (-353.68 s, Holm p = 0.0009) and spike (-73.70 s, Holm p = 0.0268), driven by the severe oversubscription cliff of 10 threads under external load, and `rule_adaptive vs balanced` in relief (-3.66 s, Holm p = 0.0177).

11.4 Reaction delay in the spike scenario

In `spike`, all three adaptive policies lose approximately 21.5-21.8 s to fixed `balanced` (37.7 s). With all contexts retained, `rl_adaptive` averaged 59.50 s (paired diff +21.80 s, 95% CI [7.58, 35.54], raw p = 0.020, Holm p = 0.44). `rule_adaptive` averaged 59.45 s (paired diff +21.75 s, raw p = 0.0057, Holm p = 0.14). The adaptive policies switch late because with a 10-token decision interval and patience 2, they require 10-15 tokens at approximately 2.5 s per token under 8 workers before leaving static.

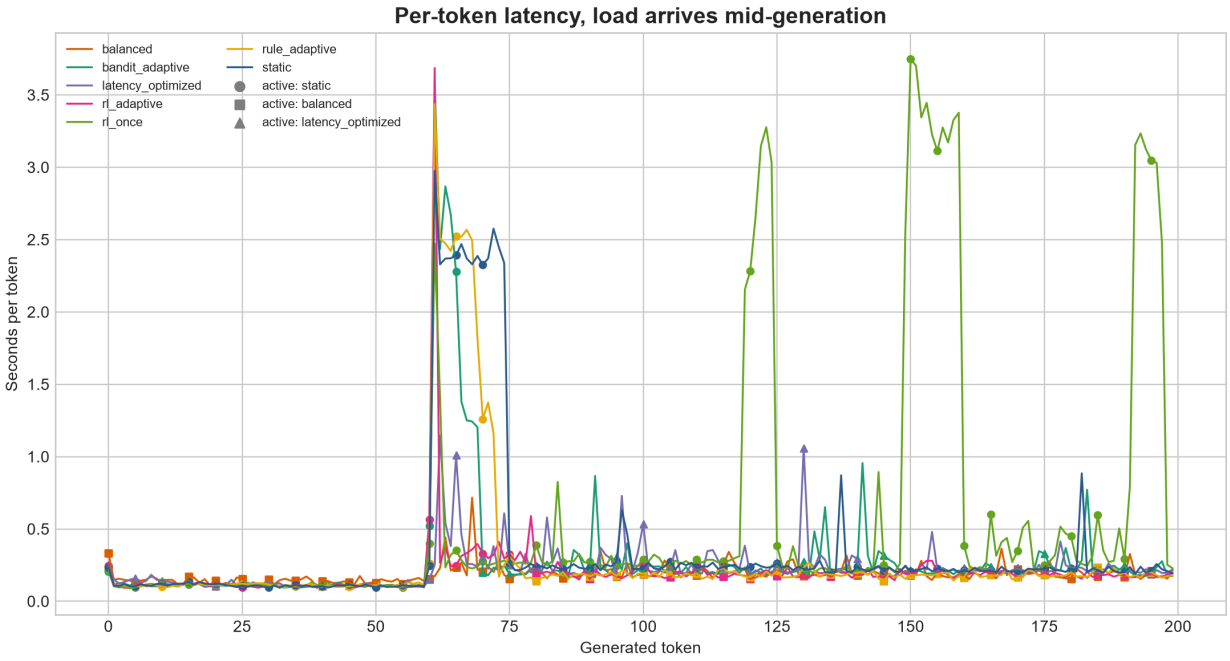


Figure 4. v5 per-token latency for one spike context. Load arrives at token 60; markers show the active strategy at each decision.

11.5 The online bandit in v5

Scenario	bandit vs best fixed	bandit vs rl_adaptive	bandit vs rule_adaptive
ramp	+43.0% vs balanced	+23.2%	+50.6%
relief	+1.6% vs latency_opt	+5.3%	+12.4%
spike	+57.0% vs balanced	-0.5%	-0.4%
steady_heavy	+24.7% vs balanced	-2.2%	+25.0%
steady_idle	+4.4% vs static	+4.2%	+5.4%

In v5 the online bandit was the slowest adaptive policy under heavy contention. In ramp context 14, the bandit explored `static` at tokens 180-200 under 12 workers, incurring 75.0 s for 20 tokens due to exploring above the oversubscription cliff. Across all contexts, bandit regret against the best fixed strategy fell from 17.16 s in the first half to 7.75 s in the second half, indicating active learning, but exploration under contention carried a steep penalty.

11.6 Sensitivity analysis: excluding latency outliers

The historical v5 report excluded four contexts (14, 19, 36, 39) using a heuristic 2x median latency rule. The source CSV contains no page-fault or swap counters to verify these as paging events. Audit of the excluded runs reveals that context 36 was an oversubscription cliff after 12 workers arrived at token 140, context 14 was a bandit exploration failure above the cliff, context 39 was a policy choice running static under 10 workers, and context 19 was an unverified suspect. As a sensitivity analysis, excluding these four contexts yields n=6-8:

Comparison (filtered sensitivity set)	Mean diff (s)	Raw p (t-test)	Holm p (30 tests)	Verdict
ramp: rl_adaptive vs rl_once	-334.6 s	0.0008	0.024	significant after Holm adjustment
ramp: rl_adaptive vs balanced	-5.1 s	0.0103	0.217	not significant
relief: rule_adaptive vs balanced	-3.5 s	0.0024	0.062	not significant
spike: rule_adaptive vs balanced	+21.7 s	0.0057	0.126	not significant
rl_adaptive vs rule_adaptive (all 5 scenarios)	-0.6 to +4.8 s	>= 0.297	1.000	not significant

Excluding contexts 14 and 36 moves the ramp comparison of `rl_adaptive` and `rl_once` in opposite directions. In context 36 `rl_adaptive` hit the cliff and `rl_once` did not (167.7 s vs 58.3 s). In context 14 `rl_adaptive` was faster (53.8 s vs 63.0 s). (Earlier versions of this report said `rl_adaptive` hit the cliff in both.) The net effect enlarges the apparent benefit of re-deciding (-334.6 s in the sensitivity set vs -238.40 s with all contexts). In all contexts, `rl_adaptive` is 16% slower than fixed `balanced` in ramp.

> Corrected reading of v5: with all 40 contexts retained and Holm adjustment over the 30 comparisons, the data does not establish that `rl_adaptive` outperforms the best fixed or rule baseline. Re-deciding DQN and the 60% threshold rule are statistically indistinguishable. The gain comes from intra-generation re-deciding, not from learning.

12. Methodology and statistics

- Paired, randomized measurements: all conditions run back to back on the same prompt under the same load scenario, in shuffled order, after a warm-up generation. This removes the run-order confound of v1/v2.
- Pre-decision state: `[prompt_length, input_tokens, cpu%, ram%]` is measured before the request, identically in training and serving.
- Fixed-length requests: v4/v5 generate exactly 200 tokens, so latency is comparable.
- Load change at a token index, not a time, so every policy faces it at the same point of the response.
- Statistics: v5r reports 95% paired bootstrap CIs, paired t-tests and Wilcoxon signed-rank tests. Holm correction spans each predeclared family across all scenarios and baselines. The legacy `significant` field aliases the adjusted paired-t test; a separate flag reports whether the bootstrap CI excludes zero. The older v3-v5 report labels used a different unadjusted rule.
- Exclusions: the original v5 prose applied a latency-only outlier rule. Because the source has no page-fault, paging, or swap counters, v5r excludes no contexts in its primary analysis and presents the four heuristic flags only as sensitivity candidates.
- Output equivalence is checked on token IDs for every context.
- Handoff timings are in-process `clone()` copies, a lower bound on a real inter-device transfer.

13. Engineering, testing and reproducibility

- Tests: the focused pipeline and adaptive policy suites cover unit logic and mock pipelines. The real-model `tests/test_functional_equivalence.py` runs the full TinyLlama-1.1B engine and verifies token-level equivalence against monolithic generate().
- `evaluate_adaptive.py` supports `--resume`, `--report-only`, configurable scenarios, decision intervals 5/10/20, patience 0/1/2, thread budgets 2/4/6/8/10/12, a decision-boundary emergency threshold, and `--keep-awake`. The new `cores_heuristic` and `rule_noh` policies are covered by deterministic tests. `segment_oracle` is an offline-only utility and was not scored on v5 because per-segment counterfactual outcomes are absent.
- `run_overnight.cmd` resumes the v5 run unattended with `HF_HUB_OFFLINE=1` and appends to a log.
- Every data version is written under its own tag (v2, pilot, v3, v4, v5). Nothing older is overwritten.
- Environment quirks: the report metadata describes a 10-physical-core, 16-logical-core, 16 GB laptop and CPU-only float32 TinyLlama. The v5 raw run CSV does not record page-fault or swap activity; high RAM snapshots and latency spikes alone do not prove paging. Analysis environment versions and collection-time metadata gaps are documented in the v5r text report.

14. Limitations and open problems

1. Small samples: v5r has 8 contexts per scenario. Several paired intervals remain wide, and selecting the best fixed mean from the same observations is descriptive rather than an independent confirmatory comparison.
2. One machine, one model, CPU only. The thread-budget regimes are specific to this 10-core / 16-thread laptop and will move on other hardware.
3. Paging: the raw v5 table lacks independent paging telemetry. Four latency-only candidates are marked in the v5r audit, not treated as verified paging events; an earlier manually removed context is not present in the supplied source CSV, so its alleged corruption cannot be independently evaluated here.
4. Synthetic load: busy-loop worker processes are a simple model of contention. Real co-located workloads also compete for memory bandwidth and cache.
5. Reaction delay: the all-context v5 mean in `spike` is descriptively higher for adaptive policies than fixed `balanced`; adjusted tests do not establish the cause or a general effect.
6. Oracle evaluation requires matched per-segment counterfactual outcomes that are not present in the current data.
7. The offline DQN goes stale when machine conditions change (seen between v3 and v4). It was trained on loads up to 8 workers only.
8. The "partitioning" is within one device. Layer stages run sequentially, so stage splits only change thread budgets, not parallelism. A CPU/GPU split (RTX 3050) has not been tried.
9. The execution graph in `src/graph` is descriptive (5 coarse nodes plus partition metadata). The scheduler does not search over graph partitions.

15. Implementation status, v6 load sweep, and deferred experiments

15.1 Specification checklist and implementation audit

Following the requirements in `Agent_Brief_v6_Improvements(2).pdf`, implementation and verification status across all P0 and P1 items are detailed below:

Requirement	Description	Status	Artifacts & Evidence
P0.1 Stall detection & exclusions	Independent paging telemetry, two-stage classification, regression test preventing cliff exclusions.	**Experimentally validated**	`src/telemetry/stall_detection.py`, `adaptive_report_v5r.txt`, `adaptive_exclusions_v5r.csv`. Unit tests pass.
P0.2 Statistical testing & Holm correction	Predefined 30-test comparison family, paired alignment, t-test, Wilcoxon, Holm p-values.	**Experimentally validated**	`src/benchmark/evaluate.py`, `adaptive_paired_diffs_v5r.csv`. Holm p = 1.0 for RL vs threshold rule.
P0.3 Reframe claims & thread budgets	Lead with best-fixed/rule comparisons, honestly report {3, 6, 10} thread budgets, sequential balanced execution.	**Experimentally validated**	Report sections 1, 3, 5, 11 updated; original project title preserved.
P1.1 Missing baselines	Implement `cores_heuristic`, `rule_noh` (patience 0), and `segment_oracle` regret bound.	**Tested** (Live benchmark **Pending**)	`src/scheduler/adaptive.py`, `src/benchmark/oracles.py`. Regression tests pass. 675-run live evaluation pending.
P1.2 Reaction-delay fix	Emergency switch on latency spike (>3x recent median) to conservative strategy; ablation grid runner.	**Tested** (Ablation run **Pending**)	`src/llm/inference.py`, `configs/config.yaml`, `run_pilot_ablation.py`. 60-run pilot ablation deferred.
P1.3 Online bandit algorithmic fixes	Drop 2 switch tokens, prune arms >2x best and top-bin static, cap prior <=75% CPU, 5-segment stall reset.	**Tested** (Live benchmark **Pending**)	`OnlineBanditPolicy` in `src/scheduler/adaptive.py`. Unit tests pass covering cost attribution, pruning, and reset.
P1.4 Expanded action space & load sweep	Single-stage budgets {2, 4, 6, 8, 10, 12}, 144-run sweep across 0-14 workers at 50 tokens each (Prompt 1).	**Experimentally validated**	`src/llm/partition.py`, `load_sweep_v6.csv` (144 runs, 0 nulls, 0 failures). Optimal budget scales down with load.
P1.4 Retraining & offline validation	Retrain contextual DQN (gamma=0, prompts 21-40 held out); offline evaluation on held-out load split (repeat 2).	**Experimentally validated** (Prompt 1) / **Pending** (Unseen prompts)	`training_data_v6.csv`, `dqn_model_v6.pth` (loss converged 0.734 to 0.056), `rl_offline_evaluation_v6.csv` (mean regret 0.0112). Generalization to unseen prompts not yet tested.
P1.5 Data scaling & bfloat16	bfloat16 equivalence validation (51.5% RAM reduction), independent paging telemetry, 15 contexts/scenario plan.	**Tested** (Benchmark **Pending**)	`test_bfloat16.py` passes (100% token match, RAM 4.4 GB to 2.1 GB). Full 14-hour benchmark (675 runs) deferred.

15.2 v6 Load sweep results (P1.4)

The 144-run full factorial load sweep was completed across 8 background worker levels (0 to 14 workers) × 3 repeats × 6 single-stage thread budgets ({2, 4, 6, 8, 10, 12} threads), evaluated at 50 tokens per run using Prompt 1 (`data/processed/load_sweep_v6.csv`). The empirical decode latency per token (seconds/token) is summarized below:

Correction (19 October 2026): the table in earlier versions of this report did not match `load_sweep_v6.csv` (40 of 48 cells differed at the printed 4 decimals, e.g. 1.2592 s/token at 8 workers / 10 threads where the CSV mean is 0.2176). It is now generated from the CSV; the fastest budget per row and the 8.4× / 12.0× ratios are unchanged.

Load workers	2 threads	4 threads	6 threads	8 threads	10 threads	12 threads	Fastest budget
0 (idle)	0.1928	0.1877	0.1737	0.1691	**0.1673**	0.1820	**10 threads**
2	0.2077	0.1830	0.1841	**0.1739**	0.1782	0.1770	**8 threads**
4	0.1987	0.1889	**0.1755**	0.1926	0.1877	0.1902	**6 threads**
6	0.1911	**0.1748**	0.1862	0.1988	0.1985	0.1896	**4 threads**
8	**0.2035**	0.2305	0.2215	0.2459	0.2176	0.2261	**2 threads**
10	**0.2421**	0.2637	0.5135	0.7101	0.5361	0.2711	**2 threads**
12	**0.3974**	0.8677	2.8918	3.1306	3.3200	3.2864	**2 threads** (8.4× faster than 10 threads)
14	**0.4088**	1.1876	4.8610	4.8122	4.8968	5.4555	**2 threads** (12.0× faster than 10 threads)

Mean decode seconds per token over 3 repeats per cell, generated from `data/processed/load_sweep_v6.csv` (144 runs) by `experiments/build_report_tables.py`.

15.3 DQN v6 training and offline validation

Contextual-bandit DQN training (gamma = 0) was executed on `data/processed/training_data_v6.csv` (144 rows, 24 paired contexts). Prompts 21-40 remained strictly held out.

- **Training trajectory (100 episodes):** Average loss converged from 0.7343 to 0.0562 (92.3% decrease), and cumulative reward improved from -12.859 to -10.420. Model weights were saved to `data/processed/dqn_model_v6.pth`.
- **Offline evaluation on held-out contexts (repeat 2, 8 contexts):** Mean regret was **0.0112**. The RL policy achieved a mean latency of **13.909 s** versus the theoretical per-context oracle of **13.629 s** (within 2.0% of the upper bound). By comparison, fixed 10 threads averaged 64.979 s, fixed 6 threads averaged 63.462 s, and fixed 2 threads averaged 14.280 s.
- **Scope and limitation:** The training and validation sets were formed across background contention conditions for **Prompt 1** (repeats 0 & 1 for training, repeat 2 for validation). Consequently, *generalization to unseen prompts has not yet been demonstrated* in this offline experiment.

15.4 Deferred experimental gates (status at 12 October; both were later run under the v7 plan, see section 16)

1. **Gate 1 (P1.2 Pilot ablation):** Script `run_pilot_ablation.py` is ready to evaluate decision interval × patience × emergency switch on `spike` and `ramp` scenarios (~55 minutes). Deferred pending explicit approval.
2. **Gate 4 (P1.5 Full v6 live evaluation benchmark):** Full comparison across 15 contexts × 5 scenarios × 9 conditions (675 requests, ~14 hours) using `evaluate_adaptive.py --tag v6 --keep-awake --resume`. Deferred pending explicit approval.

16. Phase 8 (v7): pre-registered main run

v7 replaced the three-strategy action space with single-stage thread budgets (2, 3, 4, 6, 10 threads, plus `balanced` = 3+3 sequential stages). It added per-10-token segment logging with a machine-wide paging counter (Windows "Pages Input") and a RAM guard that waits, then skips a run if RAM is above 85% before it starts. All policies were built and frozen before the main run:

- `rule_lookup` maps measured external CPU to the best arm from the v7 load sweep (bins at 63.8 / 75.1 / 83.8% to `threads_6` / `threads_4` / `balanced` / `threads_2`). Decision interval 10 and patience 1 were chosen in a reaction-delay pilot (`experiments/e2_frozen.yaml`).
- `bandit_adaptive` is the online bandit, warm-started with a prior built from the same sweep (`experiments/e4_bandit_prior.yaml`).
- `rl_adaptive` is a DQN retrained on the sweep. In leave-one-prompt-out evaluation it picked the best arm in 35% of held-out contexts (56 / 31 / 19% per fold) with mean regret 0.0053 s/token, against 79% and 0.0014 s/token for the fold-built lookup rule (`experiments/e3_dqn_v7.txt`). It stayed in the plan as a pre-registered comparison.
- `cores_heuristic` picks a thread count from the number of idle cores.

16.1 Design (experiments/e4_plan.yaml)

- 10 prompts (IDs 21-30) x 5 scenarios = 50 contexts, 9 conditions, 450 runs of 200 tokens. Condition order is randomised within each context with seed 11.
- Fixed arms: `threads_2`, `balanced` (3+3), `latency_optimized` (6), `static` (10). Adaptive: `rule_v5` (old 60% threshold), `rule_lookup`, `cores_heuristic`, `rl_adaptive`, `bandit_adaptive`.
- Primary analysis, as pre-registered: 20 paired t-tests on per-context latency (rule_lookup vs balanced and vs threads_2; rl_adaptive vs rule_lookup; cores_heuristic vs rule_lookup; each in 5 scenarios), Holm correction over all 20 at alpha 0.05. Wilcoxon signed-rank p and 95% bootstrap CIs are reported alongside, not as the decision test. With n = 10 the smallest possible two-sided Wilcoxon p is 2/1024 = 0.002, so a Wilcoxon p of 0.002 means all 10 pairs point the same way. No exclusions. Sensitivity: the same tables with paging-flagged runs removed.
- Everything else, including all comparisons with `bandit_adaptive`, `latency_optimized`, `static`, `rule_v5` and the best fixed arm, is descriptive.
- Amendment (18 Oct, before any analysis of main-run outcomes): prompts 21-30 instead of 21-35 (675 to 450 runs) because of limited machine time. Conditions, scenarios, seeds and the test family were unchanged, and the 67 runs already made matched the amended order. The resume plan had printed median latencies per condition from those 67 runs, as a time estimate, before the amendment.

16.2 Headline: rule_lookup vs the best fixed arm (descriptive)

The fairest single comparison is against the fixed arm that was fastest in each scenario, chosen afterwards from the same data. That choice favours the fixed arm, and it is not one of the pre-registered tests.

Scenario	Best fixed arm	rule_lookup (s)	Best fixed (s)	Diff (s)	%	Contexts rule_lookup faster
steady_idle	latency_optimized	16.42	16.44	-0.02	-0.1	7 / 10
steady_heavy	balanced	23.26	23.70	-0.44	-1.9	8 / 10
spike	balanced	21.25	22.33	-1.08	-4.8	10 / 10
relief	latency_optimized	18.93	19.60	-0.67	-3.4	9 / 10
ramp	balanced	22.15	22.69	-0.54	-2.4	8 / 10

From `experiments/e4_best_fixed.csv` (`experiments/analyze_main_v7.py`). Negative = rule_lookup faster.

Against the best fixed arm, rule_lookup is 0.1% to 4.8% faster. The gain is largest in spike, where load arrives mid-request, and rule_lookup is faster in all 10 spike contexts. In steady_idle it ties. The larger figures in section 16.4 (-13%, -20%) are against `balanced` and `threads_2` specifically, which are not the best fixed arm in every scenario.

16.3 Mean latency (seconds, 200 tokens, n = 10 per cell)

Scenario	threads_2	balanced	latency_opt	static	rule_v5	rule_lookup	cores_heur.	rl_adaptive	bandit
steady_idle	20.55	18.96	16.44	16.69	16.69	**16.42**	16.67	16.72	16.91
steady_heavy	25.82	23.70	25.47	30.94	24.62	**23.26**	25.57	28.12	23.49
spike	24.23	22.33	22.97	30.33	23.28	**21.25**	23.57	24.26	22.43
relief	22.53	20.77	19.60	21.15	21.97	**18.93**	20.03	20.81	19.87
ramp	24.13	22.69	24.08	90.60	40.63	22.15	32.85	35.43	**21.96**

From `experiments/e4_latency.csv`. Bold = lowest mean in the row. Means of the slow conditions in ramp are driven by single runs above the oversubscription cliff (static up to 163 s; cores_heuristic one run at 123.9 s during a paging burst).

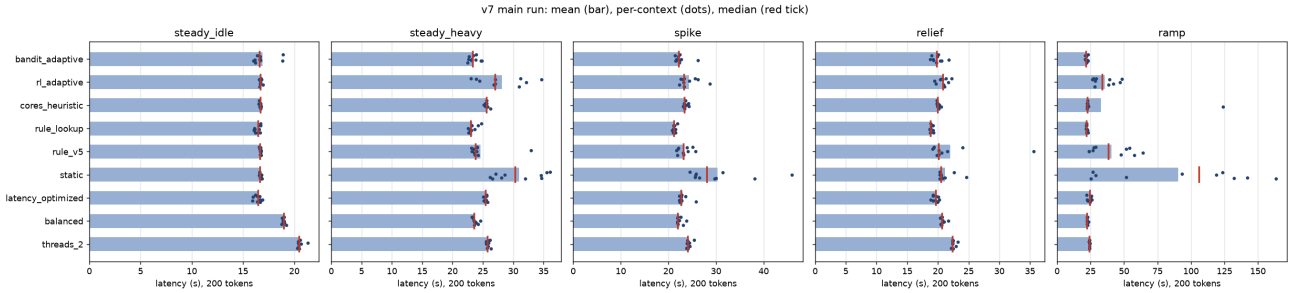


Figure 5. v7 main run: mean (bar), per-context values (dots) and median (red tick) per condition and scenario.

16.4 Pre-registered tests (a - b, negative = a faster; paired t, Holm over 20)

a vs b	Scenario	Mean diff (s)	%	95% CI (s)	Holm p (t)	Wilcoxon p (raw)	Verdict (Holm t)
rule_lookup vs balanced	steady_idle	-2.53	-13.4	[-2.69, -2.39]	<0.0001	0.002	significant
	steady_heavy	-0.44	-1.9	[-0.91, 0.07]	0.254	0.105	not significant
	spike	-1.08	-4.8	[-1.47, -0.74]	0.0038	0.002	significant
	relief	-1.84	-8.9	[-2.04, -1.65]	<0.0001	0.002	significant
	ramp	-0.54	-2.4	[-0.85, -0.20]	0.053	0.020	not significant
rule_lookup vs threads_2	steady_idle	-4.12	-20.1	[-4.30, -3.97]	<0.0001	0.002	significant
	steady_heavy	-2.56	-9.9	[-2.90, -2.16]	<0.0001	0.002	significant
	spike	-2.98	-12.3	[-3.19, -2.77]	<0.0001	0.002	significant
	relief	-3.60	-16.0	[-3.86, -3.36]	<0.0001	0.002	significant
	ramp	-1.98	-8.2	[-2.31, -1.57]	<0.0001	0.002	significant
rl_adaptive vs rule_lookup	steady_idle	+0.29	+1.8	[0.13, 0.47]	0.053	0.010	not significant
	steady_heavy	+4.85	+20.9	[2.63, 7.20]	0.021	0.004	significant (DQN slower)
	spike	+3.01	+14.1	[1.91, 4.30]	0.0088	0.002	significant (DQN slower)
	relief	+1.88	+9.9	[1.32, 2.44]	0.0016	0.002	significant (DQN slower)
	ramp	+13.28	+59.9	[8.38, 18.33]	0.0059	0.002	significant (DQN slower)
cores_heuristic vs rule_lookup	steady_idle	+0.25	+1.5	[0.11, 0.40]	0.053	0.004	not significant
	steady_heavy	+2.31	+9.9	[1.87, 2.68]	<0.0001	0.002	significant (heuristic slower)
	spike	+2.32	+10.9	[2.01, 2.66]	<0.0001	0.002	significant (heuristic slower)
	relief	+1.10	+5.8	[0.91, 1.30]	<0.0001	0.002	significant (heuristic slower)
	ramp	+10.70	+48.3	[0.37, 31.05]	0.318	0.0098	not significant

n = 10 contexts per test, from `experiments/e4_tests.csv`. Wilcoxon p-values are raw (not Holm-adjusted); the smallest possible value at n = 10 is 0.002.

15 of the 20 tests are significant after Holm correction. **cores_heuristic vs rule_lookup in ramp** is the clearest case where the t-test and the rank test disagree. The mean difference is +10.70 s (+48%), but one

context took 123.9 s during a paging burst, which widens the variance; the paired t-test gives raw p = 0.318 (Holm p = 0.318). The Wilcoxon raw p is 0.0098: cores_heuristic was slower in most ramp contexts, by much less than the mean suggests. With paging-flagged runs removed (n = 6), the difference is +0.56 s (+2.6%). By the pre-registered rule this test is not significant.

16.5 Sensitivity: paging-flagged runs removed

106 of 450 runs had at least one 10-token segment with 1000 or more system pages input (the pre-registered paging flag). Removing them leaves 3-9 complete pairs per test (`experiments/e4_tests_sensitivity.csv`). Every mean difference keeps its sign, and Holm-significant results drop from 15 to 10:

- rule_lookup vs threads_2 stays significant in four scenarios (not ramp, Holm p = 0.071).
- rule_lookup vs balanced stays significant in steady_idle and relief.
- rl_adaptive vs rule_lookup stays significant only in relief.
- cores_heuristic vs rule_lookup stays significant in steady_heavy, spike and relief.

With n this small, the sensitivity analysis checks direction, not strength.

16.6 Segment-oracle regret (descriptive)

Scenario	Oracle (s)	Best single fixed (s)	rule_lookup	bandit	cores_heur.	rl_adaptive	rule_v5
steady_idle	15.33	15.43	0.10	0.57	0.34	0.38	0.36
steady_heavy	22.11	22.36	-0.11	0.06	2.07	4.66	1.17
spike	19.96	20.92	0.09	1.23	2.31	3.09	2.05
relief	17.69	18.49	0.14	1.05	1.19	2.00	3.10
ramp	19.53	21.37	1.41	1.22	11.95	14.59	19.83

Summed token seconds; regret = policy minus a per-segment oracle over the four fixed arms (`experiments/e4_segment_oracle.csv`). The oracle only chooses among the four fixed arms, so a policy that also uses threads_4 or threads_6 can come out below it, as rule_lookup does in steady_heavy.

16.7 Reading

- **Policies built on direct measurements of this machine beat the offline DQN.** rule_lookup (a lookup table from the load sweep) and bandit_adaptive (an online bandit warm-started from the same sweep) are both close to the per-segment oracle. rule_lookup is within 0.15 s of it in four scenarios and bandit_adaptive within 1.23 s; both are within 1.41 s in ramp. In mean latency they differ by at most 1.18 s (5.6%, spike), with bandit_adaptive slightly ahead in ramp (21.96 vs 22.15 s) and rule_lookup ahead elsewhere. The bandit result is descriptive, since it was not in the test family.
- **The DQN did not generalise across prompts.** It picked the best arm in 35% of held-out contexts offline, and online it was 10% to 60% slower than rule_lookup in four of five scenarios. It also switched far more (10.3 switches per request in steady_heavy vs 2.6). This is a result about this DQN, trained on 3 sweep prompts with this state; it is not evidence that learning-based scheduling loses in general.
- Re-deciding during generation pays off most when load changes mid-request: in spike, rule_lookup beat the best fixed arm in all 10 contexts. Under steady load the best fixed arm is almost as good, provided one knows in advance which arm that is.
- Fixed `static` (10 threads) still falls off the oversubscription cliff in ramp and is the worst arm under load. On an idle machine, 6 and 10 threads are within about 2% of each other.

16.8 Run integrity and provenance

- All 450 runs present, 50 of 50 contexts complete, token IDs identical across conditions in every context, every run generated 200 tokens, fixed arms never switched, no duplicate cells

(`experiments/check_main_final.txt`).

- Two run commits: 67 rows at `ecec569` (13 Oct) and 383 rows at `0a8e3b2` (18-19 Oct). In the run code paths (`run_v7.py`, `src/`, `configs/`, the plan, frozen-rule, rule and bandit-prior files) the only difference is the 18 Oct plan amendment to `e4_plan.yaml`. From `0a8e3b2` to HEAD those paths are unchanged (`experiments/e4_supplement.txt`, part d). The DQN weights file is gitignored; its last modification (13 Oct 07:03 UTC) is before the first run.
- Speed matches across the two sessions. Fixed-arm decode s/token in steady_idle and steady_heavy agrees within about 1% for most arms; the largest gaps are latency_optimized in steady_idle (-4%) and static in steady_heavy (-6%, one 13 Oct run). Mean external CPU during runs was 37.1% on 13 Oct and 40.7% on 18-19 Oct overall, and within 1 point per scenario (`experiments/e4_by_date.csv`). The 13 Oct rows cover only prompts 21-22, so this check mixes session with prompt.
- `tree_dirty` is False on the 13 Oct rows and True on the 383 later rows. Git status at run start is not recorded. The flag counts the tracked output CSVs the run writes and the two tracked check reports that the pre-run gates rewrite; no code file changed during the run.
- RAM before runs: min 79.8%, median 82.7%, max 85.0%. The guard skipped 17 runs (`skipped_runs_v7.csv`); all 17 were run later with `--resume`. The run process was restarted once on 19 Oct during a guard wait; the run in progress had not been written and was redone.

16.9 Threats to validity

- **Power settings changed between sessions.** On 17 Oct the laptop gave only about 37% CPU to 16 busy workers. Changing core parking alone did not fix it; several AC core-parking and scheduling settings were then changed and python was exempted from power throttling before the 18-19 Oct session, so the 67 runs from 13 Oct and the 383 later runs were made under different power settings. The settings in force on 13 Oct were not recorded; the 17 Oct values before the change are in `experiments/powercfg_before.txt`. Gates guarded each resume: CPU capacity at 100% with 16 workers, and threads_2 and balanced within 25% of the sweep. The by-session comparison (16.8) shows fixed-arm speed within about 1% for most arms, but it rests on n = 1-2 runs per arm from 13 Oct.
- **Uneven timing within contexts.** Pairing assumes all conditions of a context run close together. Context 8 spans 13 and 18 Oct. Context 9 (prompt 22, relief) ran 3 conditions on 18 Oct at 23:59 and 6 on 19 Oct at 08:44. Contexts 46-48 each span up to 2 hours because of RAM-guard skips.
- **Memory pressure.** With the fp32 model loaded, RAM sat at 80-86%, and 106 of 450 runs had a paging event. The sensitivity analysis keeps every direction, but at n = 3-9.
- **Plan amendment after the start.** The amendment to prompts 21-30 was made after 67 runs, and after median latencies from those runs had been printed for a time estimate. It changed only the number of prompts.
- **Best-fixed comparison is chosen in hindsight** from the same data. That favours the fixed arm, and it is descriptive.
- **Scope.** One laptop (10 physical / 16 logical cores, 16 GB), one 1.1B model in fp32 on CPU, synthetic busy-loop load, 10 prompts. rule_lookup's arm map was measured on this machine, which is part of why it does well here; a different machine needs a new sweep.

17. Related work

This section covers only the six papers listed in section 18, each opened and checked (`experiments/reference_check.md`). It is not a survey, and it makes no claim about what other work exists.

Choosing the number of threads at run time. Curtis-Maury et al. [4] predict an energy-efficient level of concurrency for each phase of a multithreaded scientific program. They use multivariate regression on hardware event counters and throttle concurrency at run time. Pusukuri et al. [5] (Thread Reinforcer) find a

good thread count by monitoring a program at OS level, without recompiling it or changing OS policies. On a 24-core server they found that the thread count giving the best speedup varied widely across PARSEC programs (16 to 63 threads). Both adapt a program's thread count to that program's own scalability. This project instead adapts the thread budget of one LLM request to load from other processes on a shared 16-thread machine, re-deciding every 10 tokens.

Run-time adaptation of inference. EdgeFM [6] switches at run time between an on-device model and a cloud foundation model, based on data uncertainty and network conditions. Here the model and its output stay fixed (token IDs are identical across arms), and only the CPU thread budget changes.

Learning-based policies. `rl_adaptive` uses a deep Q-network [2] as a contextual bandit (discount 0, one decision per segment). `bandit_adaptive` is a discounted contextual bandit over CPU-load bins with a UCB-style exploration bonus, in the family of contextual-bandit methods such as [3], though it keeps per-bin estimates instead of the linear model of [3]. The model under test is TinyLlama-1.1B [1].

18. References

1. P. Zhang, G. Zeng, T. Wang, W. Lu. TinyLlama: An Open-Source Small Language Model. arXiv:2401.02385, 2024.
2. V. Mnih, K. Kavukcuoglu, D. Silver, et al. Human-level control through deep reinforcement learning. *Nature* 518(7540):529-533, 2015. doi:10.1038/nature14236.
3. L. Li, W. Chu, J. Langford, R. E. Schapire. A Contextual-Bandit Approach to Personalized News Article Recommendation. In *Proc. 19th International Conference on World Wide Web (WWW 2010)*, pp. 661-670, 2010. doi:10.1145/1772690.1772758; arXiv:1003.0146.
4. M. Curtis-Maury, F. Blagojevic, C. D. Antonopoulos, D. S. Nikolopoulos. Prediction-Based Power-Performance Adaptation of Multithreaded Scientific Codes. *IEEE Transactions on Parallel and Distributed Systems* 19(10):1396-1410, 2008. doi:10.1109/TPDS.2007.70804.
5. K. K. Pusukuri, R. Gupta, L. N. Bhuyan. Thread Reinforcer: Dynamically Determining Number of Threads via OS Level Monitoring. In *2011 IEEE International Symposium on Workload Characterization (IISWC)*, pp. 116-125, 2011. doi:10.1109/IISWC.2011.6114208.
6. B. Yang, L. He, N. Ling, Z. Yan, G. Xing, X. Shuai, X. Ren, X. Jiang. EdgeFM: Leveraging Foundation Model for Open-set Learning on the Edge. In *Proc. ACM SenSys 2023*. arXiv:2311.10986.

19. Appendix: files, commands and data index

19.1 Repository layout

```
configs/config.yaml            model + strategy definitions
src/config.py                  config loader
src/data/                      Alpaca download, preprocessing, prompt loading
src/llm/                       model loading, partition strategies, layer-sliced inference
src/graph/execution_graph.py   networkx execution graph annotated with partitions
src/telemetry/                 CPU/RAM metrics, external-load monitor, load generators
src/rl/                        features, reward, offline bandit environment, DQN agent, training
src/scheduler/                 fixed, RL, adaptive and core-heuristic policies
src/benchmark/                 paired statistics, plots and offline oracle utilities
evaluate_fixed_strategies.py   paired fixed-strategy dataset (v3, pilot)
evaluate_rl_policy.py          online evaluation of the trained DQN
evaluate_adaptive.py           intra-generation re-scheduling and v6 comparison policies
reanalyze_adaptive_v5r.py      reproducible all-context v5 reanalysis
pilot_load_sweep.py            per-strategy decode speed at 0-14 busy workers
run_overnight.cmd              historical unattended v5 run command
tests/                         pipeline, adaptive and model-equivalence tests
docs/report/                   this HTML report source
```

19.2 Reproducing the results

```
python -m src.data.download_dataset
python -m src.data.preprocess_dataset
python -m pytest tests/test_functional_equivalence.py -v
python evaluate_fixed_strategies.py --num-prompts 10 --repeats 1 --loads 0,2,4,6,8 --max-new-tokens 50
python -m src.rl.train --csv data/processed/fixed_strategy_metrics_v3.csv --tag v3
python pilot_load_sweep.py
python evaluate_adaptive.py --tag v4 --num-prompts 10 --prompt-offset 20 --max-new-tokens 200
python evaluate_adaptive.py --tag v5 --num-prompts 8 --prompt-offset 20 --heavy 10 --switch-token 60 \
    --ramp "60:8,140:12" --threshold 60 --bandit-edges "51,63" --bin-margin 4 --resume --keep-awake
python evaluate_adaptive.py --tag v5 --report-only
python reanalyze_adaptive_v5r.py
python -m pytest tests/test_pipeline.py tests/test_adaptive.py
```

> The fixed-strategy, load-sweep, adaptive v4/v5, and v6 commands above launch model workloads and can run for hours. Run them only after explicit approval. Do not reuse v5 tags for new measurements. No benchmark or model retraining was launched for this report update.

Approved v6 comparison command

```
python evaluate_adaptive.py --tag v6 --num-prompts 8 --prompt-offset 20 --heavy 10 \
  --switch-token 60 --ramp "60:8,140:12" --interval 10 --patience 2 \
  --thread-budgets "2,4,6,8,10,12" --emergency-threshold 85
```

v7 main run

```
run_main_v7.cmd [22,23,24]           machine, CPU-capacity and sanity gates, then run_v7.py --resume
python experiments/check_main_partial.py v7
python experiments/analyze_v7.py     writes experiments/e4_*.csv, e4_summary.txt, results/v7/*.png
```

19.3 Data index (data/processed)

Files	Contents
`prompts.csv`, `inference_results.csv`, `runtime_metrics.csv`	v1 prompts, inference and telemetry
`fixed_strategy_*_v2.csv`, `rl_*_v2.csv`, `dqn_model_v2.pth`	v1/v2 results (noise, reference only)
`fixed_strategy_*_pilot.csv`	Pilot: 3 prompts x 0/4/8 workers
`fixed_strategy_*_v3.csv`	v3 paired dataset, stage-level metrics, paired diffs
`rl_training_history_v3.csv`, `rl_offline_evaluation_v3.csv`, `dqn_model_v3.pth`	DQN training and held-out evaluation
`adaptive_*_v3.csv`	First re-scheduling run (60 tokens)
`adaptive_*_v4.csv`, `adaptive_report_v4.txt`	v4 runs, decisions, per-token timelines, bandit history, summary, paired diffs
`load_sweep_v4.csv`	Seconds per token per strategy at 0-14 workers
`adaptive_*_v5.csv`, `adaptive_report_v5.txt`, `adaptive_bandit_state_v5.json`	v5 runs with the ramp scenario
`adaptive_exclusions_v5r.csv`, `adaptive_summary_v5r.csv`, `adaptive_paired_diffs_v5r.csv`, `adaptive_report_v5r.txt`	all-context primary reanalysis; no paging-based exclusions
`adaptive_summary_sensitivity_v5r.csv`, `adaptive_paired_diffs_sensitivity_v5r.csv`	separate sensitivity excluding four latency-only heuristic candidates
`load_sweep_v7.csv`	v7 load sweep (288 runs), source of `experiments/e3_rule.yaml`
`v7runs_v7.csv`, `v7timelines_v7.csv`, `v7segments_v7.csv`, `v7decisions_v7.csv`, `skipped_runs_v7.csv`	v7 main run: 450 runs, per-token timelines, per-10-token segments with paging counters, policy decisions, RAM-guard skips (all later re-run)
`experiments/e4_*.csv`, `experiments/e4_summary.txt`, `experiments/check_main_final.txt`	v7 analysis outputs and final integrity checks

Plots for every version are in `results/` (latency by load or scenario, paired differences, spike and relief timelines).